MAE 409: Special Topics – Finite Element Methods
Final Project

CFD, Stress, and Deformation Analysis of a Simple Sounding Rocket

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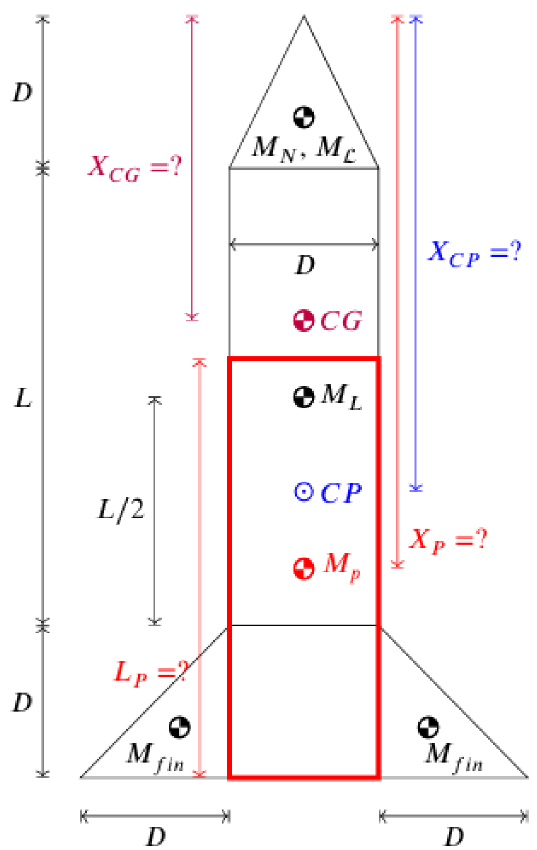
Background

A stress analysis of a simple sounding rocket was done for this project. The rocket's optimized design was a result of the final project for "MAE423: Introduction to Propulsion". However, at the end of it, there was no structural analysis required to see if the optimized design of the rocket was even viable to be used in the real world scenario if the proposed structure was not considered to be rigid. Doing this analysis would increase the understanding of how different forces and pressures need to withstood by the structure of the rocket in order to be flown completely intact.

The proposed rocket design was converged from a MATLAB script (Appendix A) that iterated L and D parameters of the rocket's dimensions along with a combination of max altitude (h_{max}), max normalized acceleration (a_{max}), and payload weight (W_L). The results for the base case (all medium options) was used because that way, only one variable would be changed in the rest of the six scenarios (if one option was selected, the rest of the options had to be a medium option according to the project guidelines). The rocket's dimensional parameters and design inputs are shown in Fig. (1) below.

Parameterized Rocket Geometry



Design Input Parameters

Design Spec.	Value	Description
$M_L(kg)$	5, 10, 20	Payload mass
$h_{max}(ft)$	$10k$, $20k$, $30k$	Maximum altitude
a^*_{max}	5, 10, 20	Normalized max acceleration
SM	2	static margin $= (X_{CP} - X_{CG})/D$
$\rho_s(kg/m^3)$	2700	shell density (aluminum)
$\rho_p(kg/m^3)$	1772	propellant density, $66-78\%$ AP, 18% organic polymer, $4-20\%$ Al
$\sigma_s(MPa)$	150	shell working stress
N	4	number of fins

Figure 1: Rocket Dimensional Diagram & Design Input Parameters from MAE423 project.

Formulation

MATLAB & OpenRocket

The results from the MATLAB script were compiled, and the parameters for the base case were selected (highlighted in Table 1) and used to model the rocket in a simulating software called OpenRocket. After modelling an appropriate motor, the rocket was "flown" vertically in standard atmospheric conditions with no wind and launch angle. Although the MATLAB script set the h_{max} to 6097 meters, the maximum apogee of the rocket when simulated in OpenRocket was around 2150 meters (Figure 2) as it took atmospheric conditions and drag factors into account.

The MATLAB script resulted in L = 2.57151m, D = 0.19424m, and a wall thickness of about 0.0837mm. OpenRocket simulation generated the results such that the maximum speed achieved by the rocket was about 236 m/s. These results were used to model two rockets in SolidWorks (one solid and the other hollow), and the generated .STEP files were imported into ANSYS Fluent and Static Structural for further analysis respectively.

Table 1: MATLAB Script Results.

	R_{opt}	W_{eq} (m/s)	T_{burn} (s)	P_o/P_a	δ/D	D (m)	L/D	X_{CG} (m)	X_{CP} (m)	M_p (kg)	M_s (kg)	M_o (kg)	λ_{max}	ε
Case 1	11	207.7025	19.2478	1.2756	4.31E-04	0.4981	9.33333	5.15481	4.15861	102.067	5.20669	112.274	0.04661	0.04854
Case 2	11	207.7025	19.2478	1.2756	4.31E-04	0.26625	6.31856	1.08389	1.61639	205.873	0.58729	226.46	0.09687	0.00284
Case 3	11	146.8679	13.6102	1.13182	3.82E-04	0.18564	17.5808	2.33551	2.70678	104.102	0.41023	114.513	0.09568	0.00393
Case 4	11	254.3826	23.5736	1.43198	4.84E-04	0.20984	6.91611	0.94899	1.36868	103.454	0.3454	113.799	0.09634	0.00333
Case 5	6	304.0674	25.8297	1.65268	5.58E-04	0.17924	1.85268	0.1248	0.48328	50.5518	0.11037	60.6622	0.19739	0.00218
Case 6	21	153.3638	14.889	1.14431	3.86E-04	0.22364	20.0089	3.22402	3.67131	216.286	0.81428	227.1	0.04606	0.00375
Case 7	11	207.7025	19.2478	1.2756	4.31E-04	0.19424	13.2389	1.80645	2.19493	104.134	0.41336	114.547	0.09565	0.00395

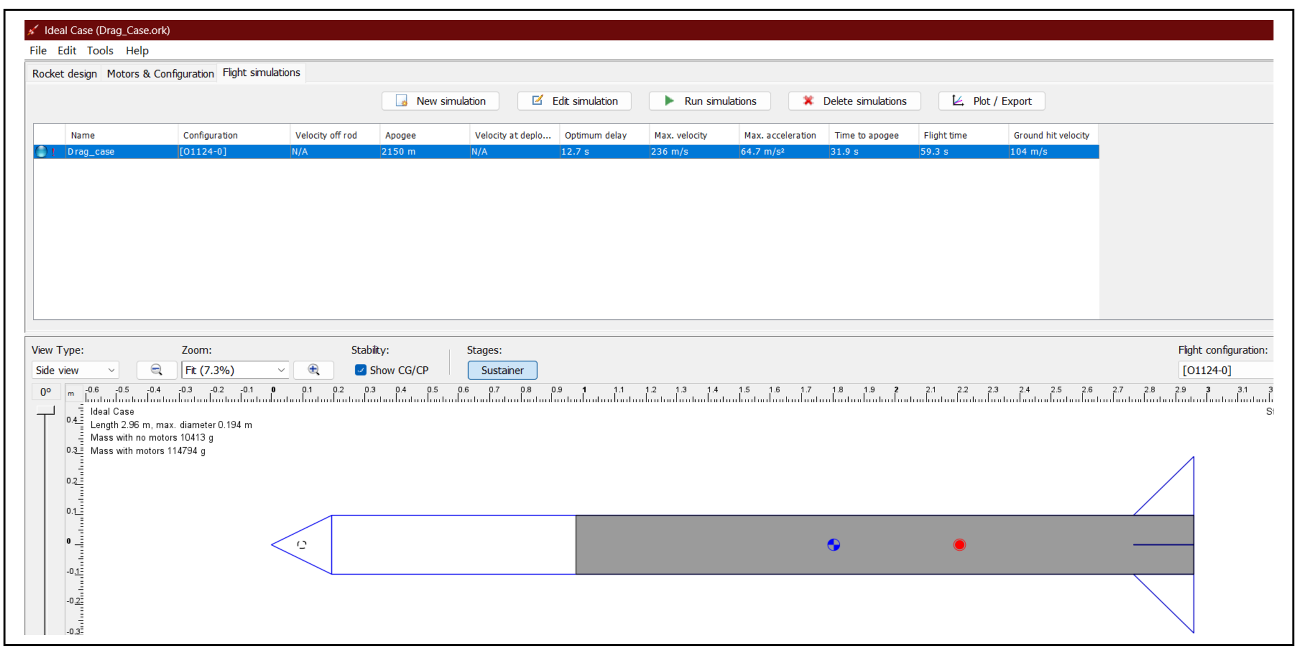


Figure 2: OpenRocket Simulation Results

ANSYS Fluid Flow – Fluent (CFD)

The maximum pressure faced by the rocket's nosecone travelling at the speed that was calculated by OpenRocket was need. In order to calculate that required pressure, ANSYS Fluent was used.

The solid rocket .STEP file was imported to Fluent on Workbench in geometry section. For the fluid flow analysis, an enclosure around the rocket body is needed which simulates a virtual wind tunnel. In order to do so, a box (10m x 3m x 3m) was created and the rocket body was centered inside it. Then, a Boolean feature was generated (in the subtract setting) whose target body was the box, and the tool body was the rocket. This allowed for the generation of an enclosure with the rocket shaped cavity inside it, and the rest of the material was set as a fluid in the material settings. The enclosure is shown in Figure 3 below.

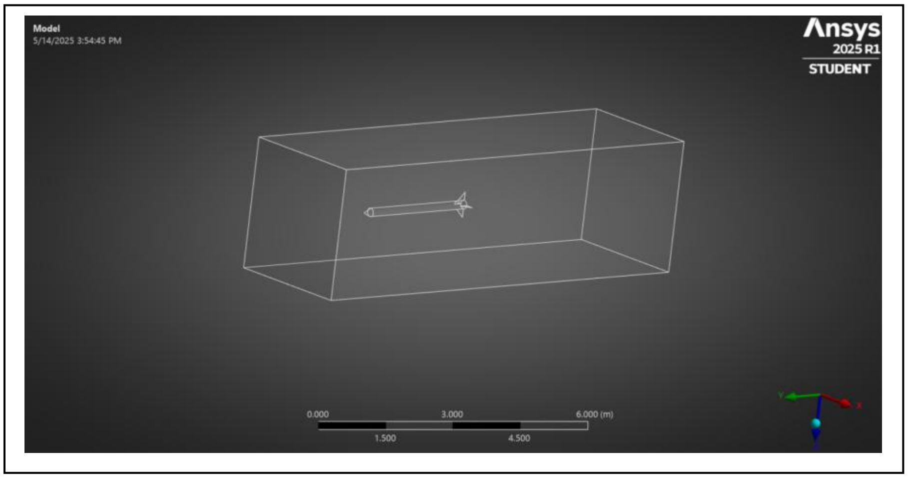


Figure 3: Enclosure with the Rocket inside.

After the enclosure was generated, the next step was to generate an appropriate mesh for the analysis. The walls of the enclosure were named according to their functions. This meant that the wall near the nosecone was renamed as inlet, the wall facing the end of the rocket was named as the outlet, and the rocket geometry was distinguished with its own unique name. The size of the mesh was set at 0.07m (resulting in total elements to be around 390,000), and the optimized option was selected. The resulting mesh was generated as shown in Figure 4.

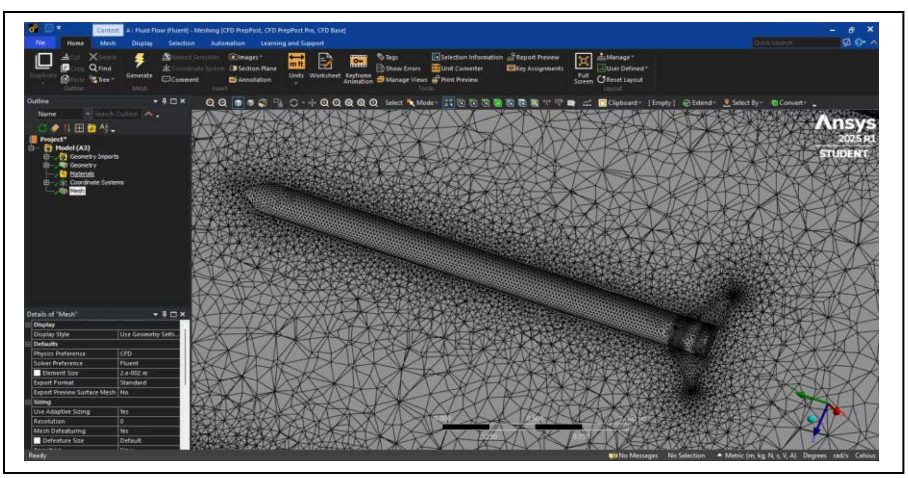


Figure 4: Generated Mesh (Spliced cross section).

After the mesh generation, the next step involved setting up the whole scenario in ANSYS Fluent (setup section). The inlet and outlet were selected as a mass flow inlet and outlet. The mass flow was calculated as shown in Eq. (1). The inlet and outlet are shown in Figure 5.

The material and fluid properties were verified in this section as well. The iteration count was set at 110 for the solution in order to have ample resolution for the results processed in the Post CFD software in the "Results" section of Fluent on Workbench. The notable result of the max pressure on the nosecone was found to be 21,480 Pa, which was used as the input for the structural analysis (shown in Figure 6).

$$\dot{m} = \rho V A_{CS} = 1.225 \times 236 \times (3 \times 3) = 2601.9\, {}^{kg}/_{s} \qquad \textbf{Eq. (1)}$$

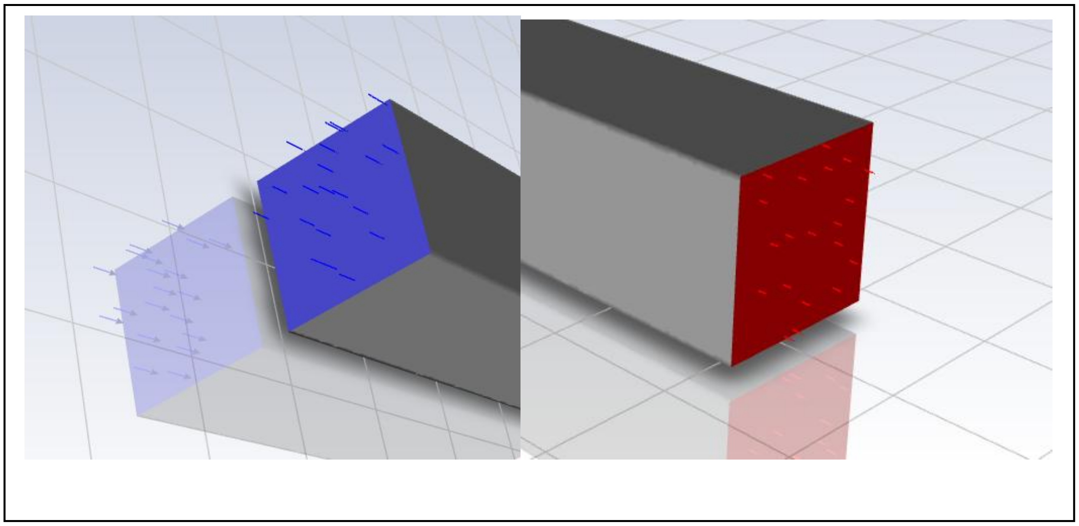


Figure 5: Defined Inlet (blue) and Outlet (red).

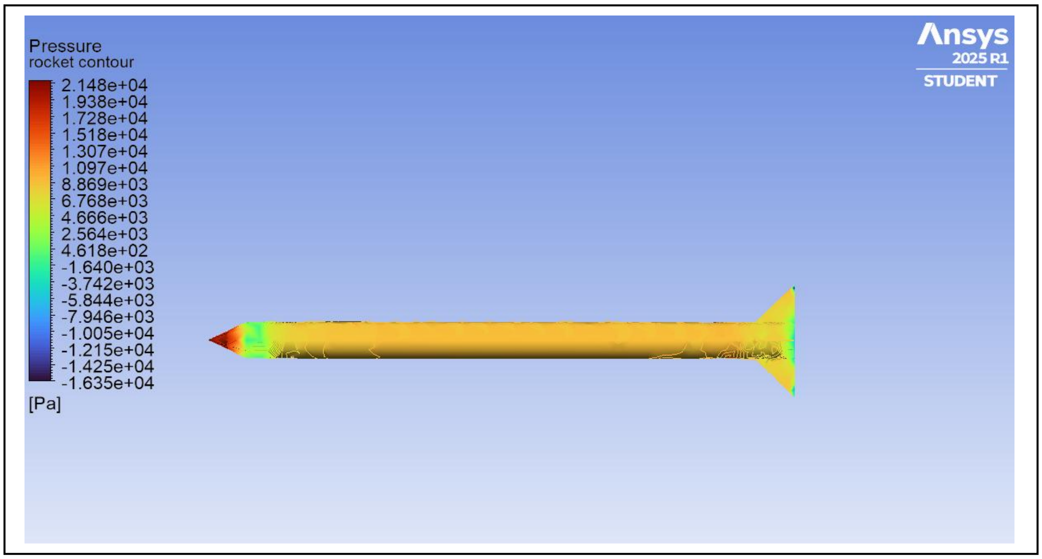


Figure 6: Pressure Contour generated from ANSYS Fluent.

Next, a pressure gradient and flow stream line were added to the analysis in order to verify whether the pressure contour was accurate for the simulation or not. The resulting pressure gradient and airflow streamlines are shown in Figures 7 and 8 respectively. It is to be noted that since the enclosure was considerably larger than the rocket, the streamlines did not have enough resolution (point count set at 50 to conserve computation power) to be shown how they go around the rocket. It does show that the air flow is simulating as intended inside the enclosure. The pressure gradient in Figure 7 also complements the pressure contour shown in Figure 6 earlier.

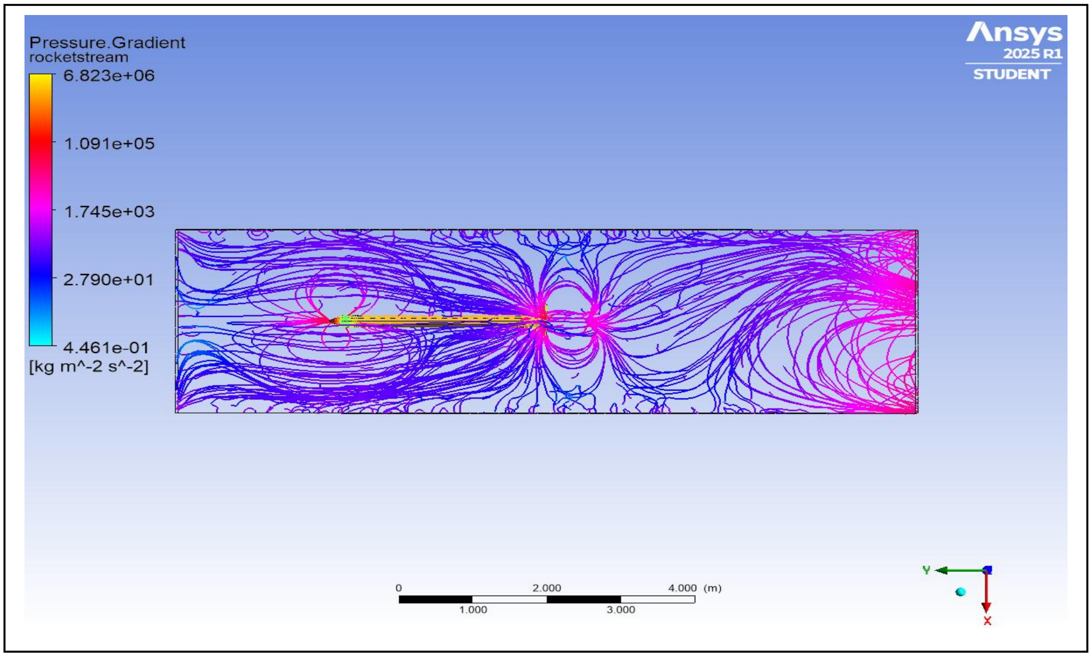


Figure 7: Pressure Gadient Lines Along the Rocket.

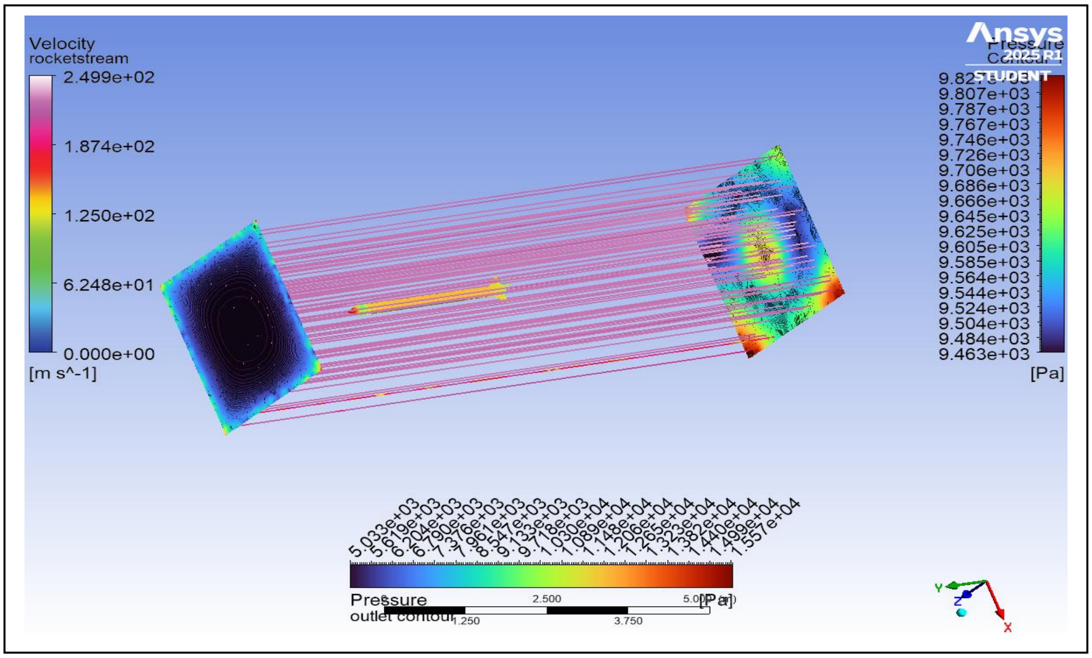


Figure 8: Airflow Stream Lines in the Enclosure.

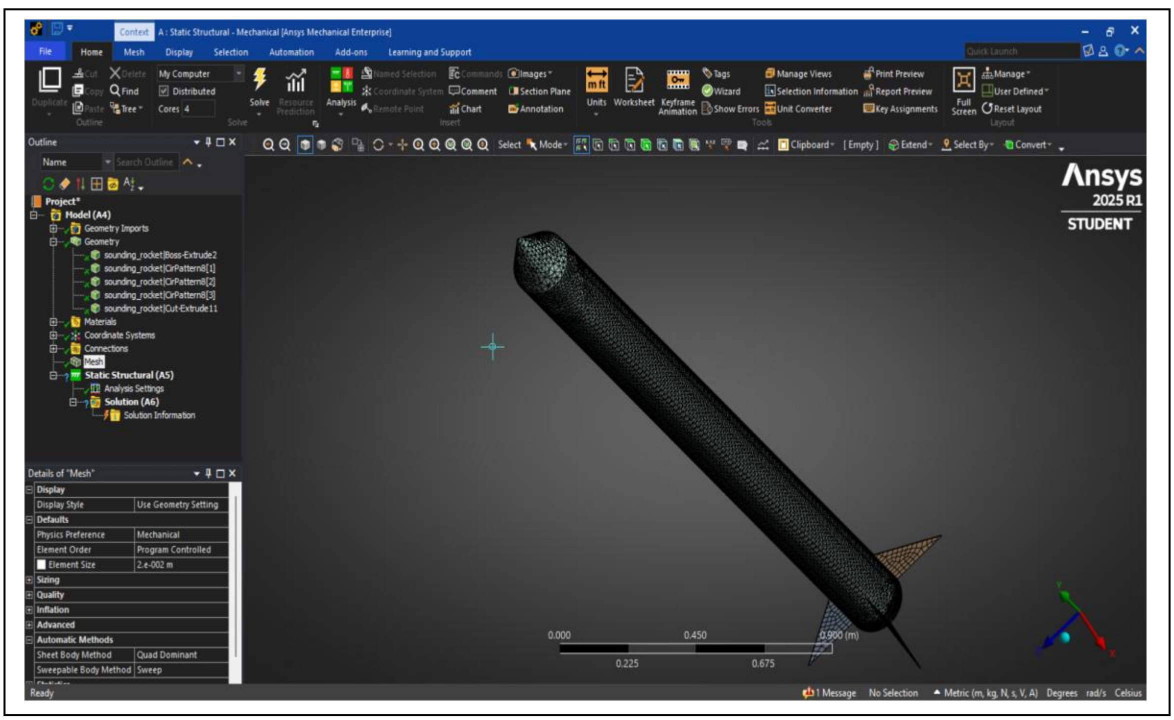
ANSYS Static Structural

After the maximum pressure on the nosecone is obtained from Fluent, the hollow rocket .STEP file is imported in the geometry section of Static Structural Workbench. It is important to note that the thickness obtained from the MATLAB script could not be implemented when using the hollow rocket because 0.0837mm was too thin for the mesh to be generated adequately due to computational and software limitations. Even a mesh size of 0.02m, was resulting in an error because wall thickness was very small.

In order to generate the mesh, the wall thickness of the cylindrical tube, nosecone, and the fins had to be increased to 0.2mm, which is a thickness increase of roughly 138.95%. Although this will impact how the rocket structure is 'fortified', we can still get a good estimate of how the thickness and material influences the stress experience by the rocket.

Once the hollow rocket's geometry was imported, the mesh was generated in the DesignModeler with the size of 0.02m as shown in Figure 9. Following this was the "Setup" phase, where the applied forces and boundary conditions were applied. The bottom ring of the cylindrical tube was set as a fixed support (Figure 10), while the obtained maximum pressure from Fluent was applied on the on the nosecone of the rocket (Figure 11).

The results needed to be observed were total deformation, equivalent stress, and equivalent strain over rockets made from 3 different materials and will be discussed in the upcoming "Discussion" section.



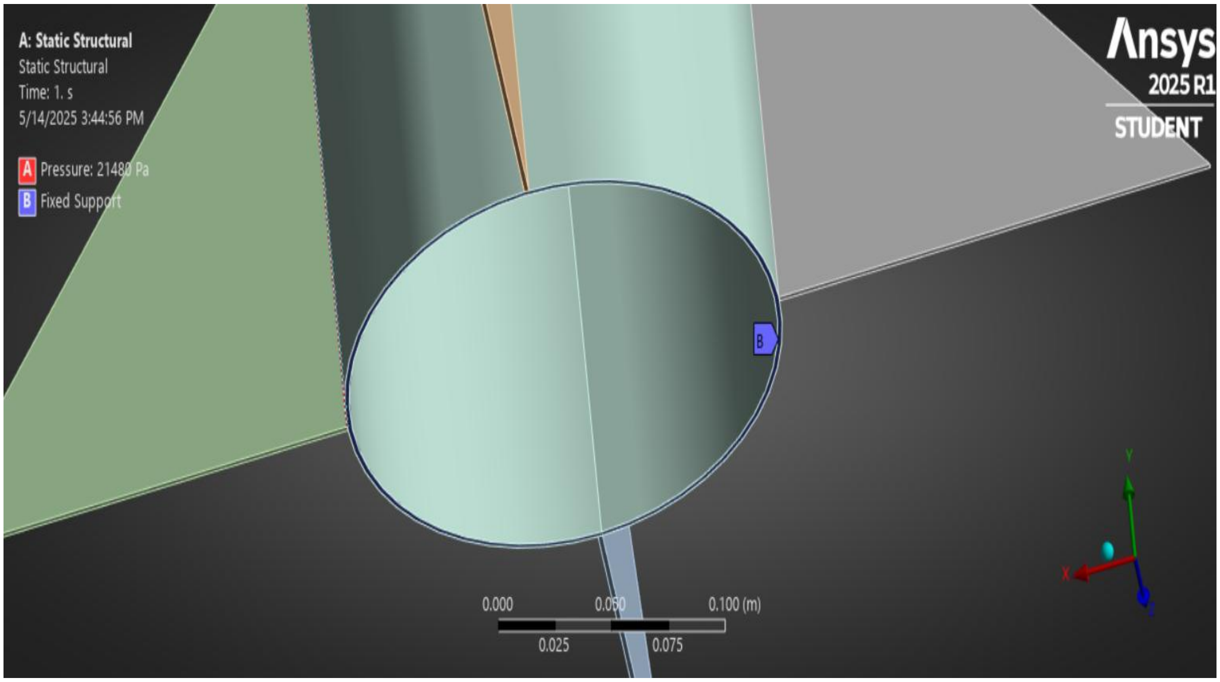


Figure 10: Fixed Support

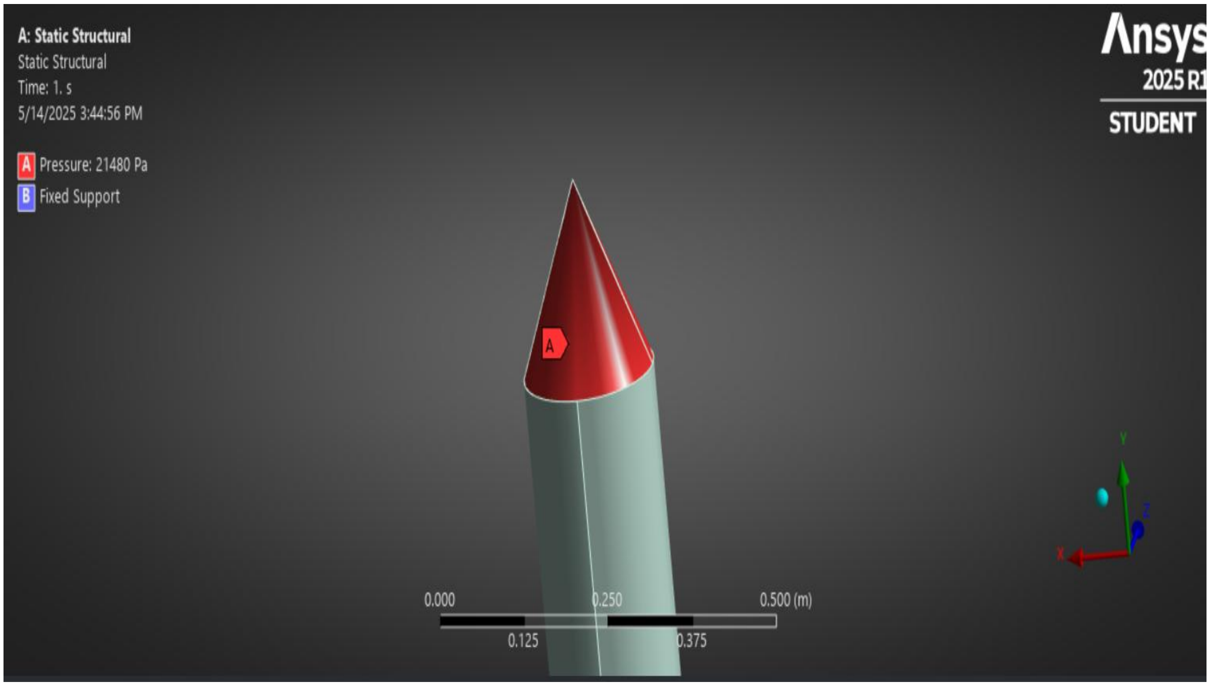


Figure 11: Applied Pressure (image is slightly warped due to fitting)

Figure 9: Generated Mesh for Structural Analysis

Results

MATLAB & OpenRocket

The results of this section are described in detail in the earlier section under the title of "MATLAB & OpenRocket". They are displayed in Table 1 and Figure 2 along with their descriptions. To reiterate, The MATLAB script resulted in L = 2.57151m, D = 0.19424m, and a wall thickness of about 0.0837mm, while OpenRocket simulated a maximum speed of 236 m/s.

ANSYS Fluid Flow – Fluent (CFD)

Similarly, the results of this section are described in detail in the earlier section under the title of "ANSYS Fluid Flow – Fluent (CFD). They are shown in Figures 6, 7, and 8. However, it is important to mention that the pressure contour of the rocket (Figure 6) was verified by the pressure gradient lines in the air around the rocket (Figure 7). The turbulence probability was set around 15% (default) in the settings that yielded the results and the truncation error tolerance (numerical error) was set at 0.01 (default). Everything that was plotted was derived based on these tolerances.

In Figure 6, it can be observed that the nosecone of the rocket experiences the most pressure (as was expected). Figure 7 shows pressure gradients stream lines separating from the nose cones that indicate that high pressure. Furthermore, the green (low pressure zone) area in Figure 6 is indicated by the "circulating" streamlines in Figure 7, confirming the low pressure zone. Figure 8, shows that the air in enclosure is flowing as intended.

ANSYS Static Structural

After the boundary conditions were set, only three different solution were called (total deformation, equivalent stress, and equivalent strain). Furthermore, three different materials were also used on the same geometries and conditions in order to understand how different materials properties would influence the obtained results. These results are progressed below.

1. Aluminum Alloy

This material was the original material from the initial project guidelines from MAE423, and the aluminum alloy that had the closest matching density was selected. The maximum deformation (Figure 12), stress (Figure 13), and strain (Figure 14) faced by the aluminum structure was 2.1537e-5 meters, 1.4186e6 Pascals, and 2.2045e-5 respectively, while the minimum values were very insignificant (compared to the maximum values) or completely 0. The deformation shown on the figures are set to auto-scale, so they seem to be visually exaggerated. This setting was chosen to have visually stimulating figures that emphasize on the nature of the shape change of rocket components under applied conditions.

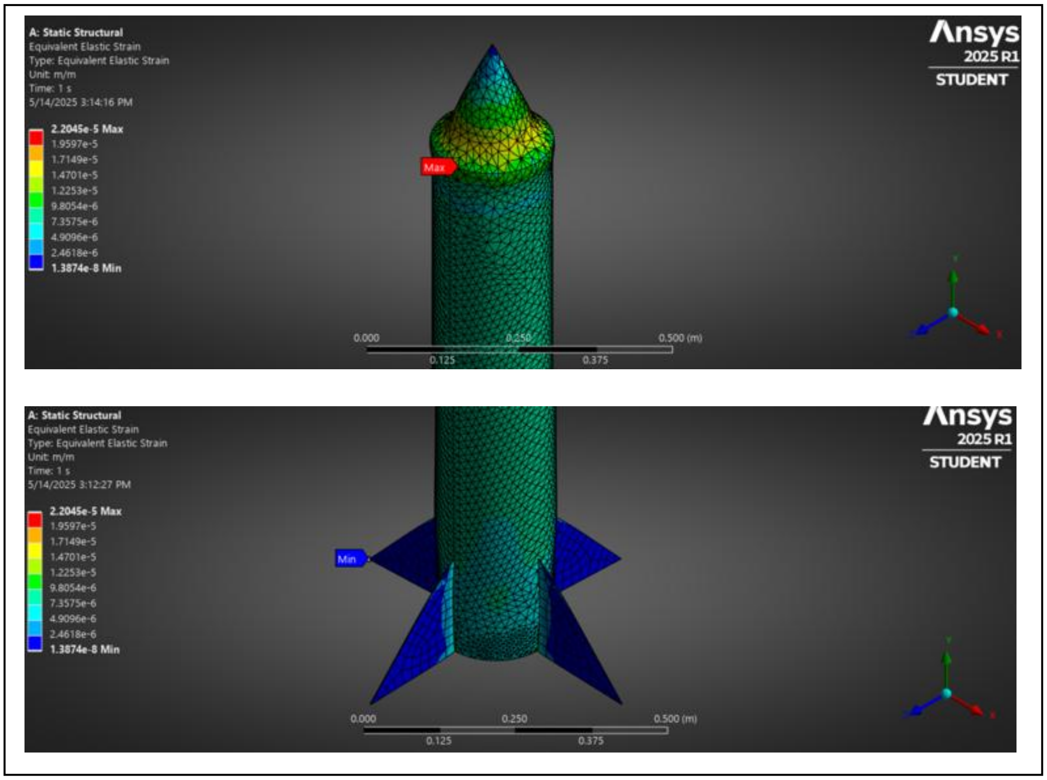


Figure 13: Max (top) and Min (bottom) Deformation of the Aluminum Rocket

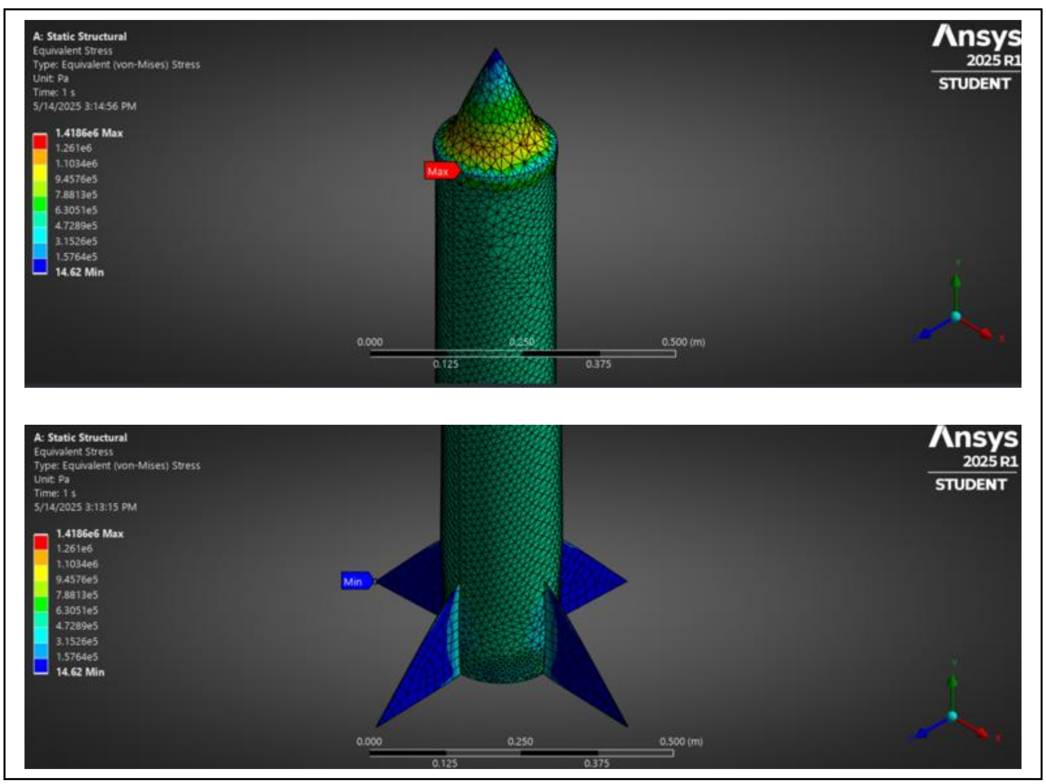


Figure 14: Max (top) and Min (bottom) Stress of the Aluminum Rocket

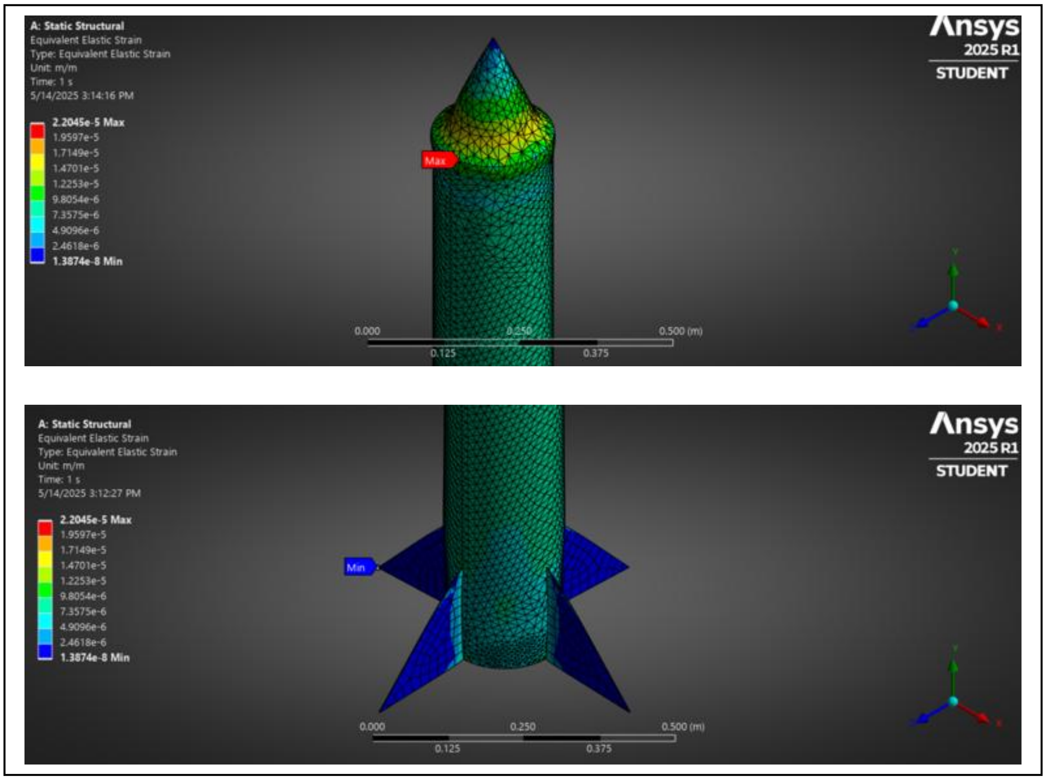


Figure 15: Max (top) and Min (bottom) Strain of the Aluminum Rocket

2. Titanium Alloy

Titanium was selected due it's obvious higher strength and hardness properties as compared to aluminum. The same conditions were applied as before, its just that material properties were swapped out in Mechanical. The maximum deformation (Figure 16), stress (Figure 17), and strain (Figure 18) faced by the titanium structure was 1.5914e-5 meters, 1.4135e6 Pascals, and 1.621e-5 respectively, while the minimum values were very insignificant (compared to the maximum values) or completely 0. Again, the indicated deformations were exaggerated due to auto-scaling due to aforementioned reasons.

The resulting deformation was 26% less than the original deformation of aluminum structure. This is due to titanium being a significantly stiffer and harder metal as compared to aluminum. It is also to be noted that titanium is also a denser material with its density measured to be 4540 kg/m^3.

A rocket with a titanium structure would be a lot stronger than its identical aluminum counterpart. However, it will also weight a lot more in comparison due to the increase in density.

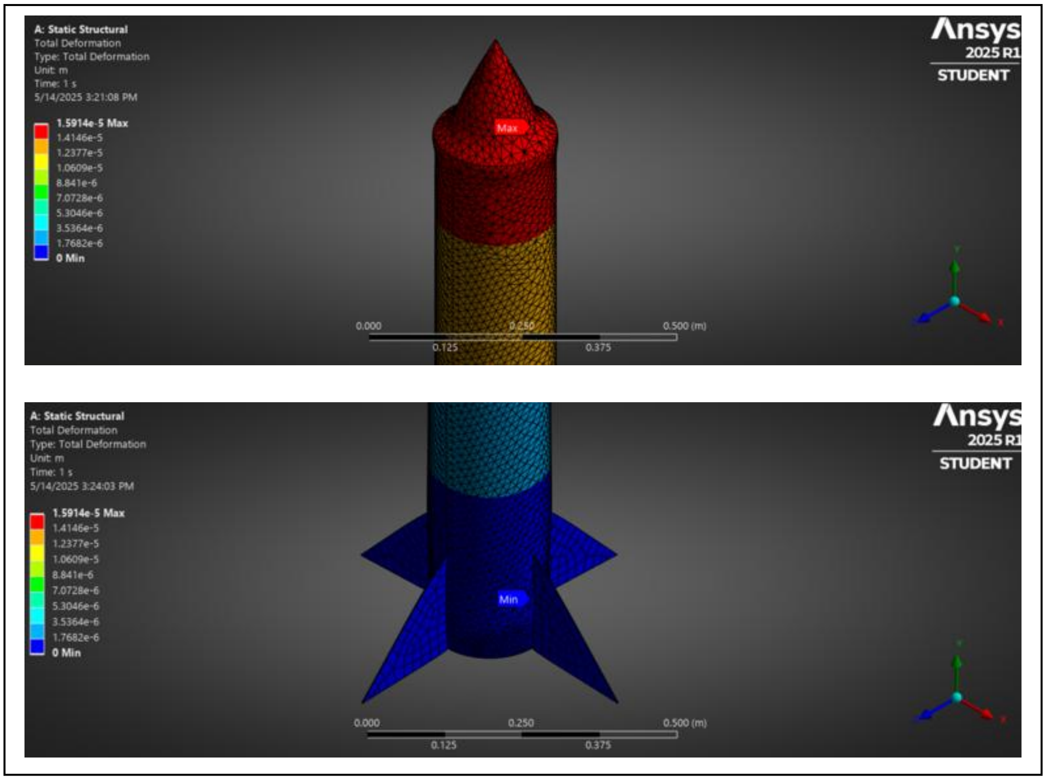


Figure 16: Max (top) and Min (bottom) Deformation of the Titanium Rocket

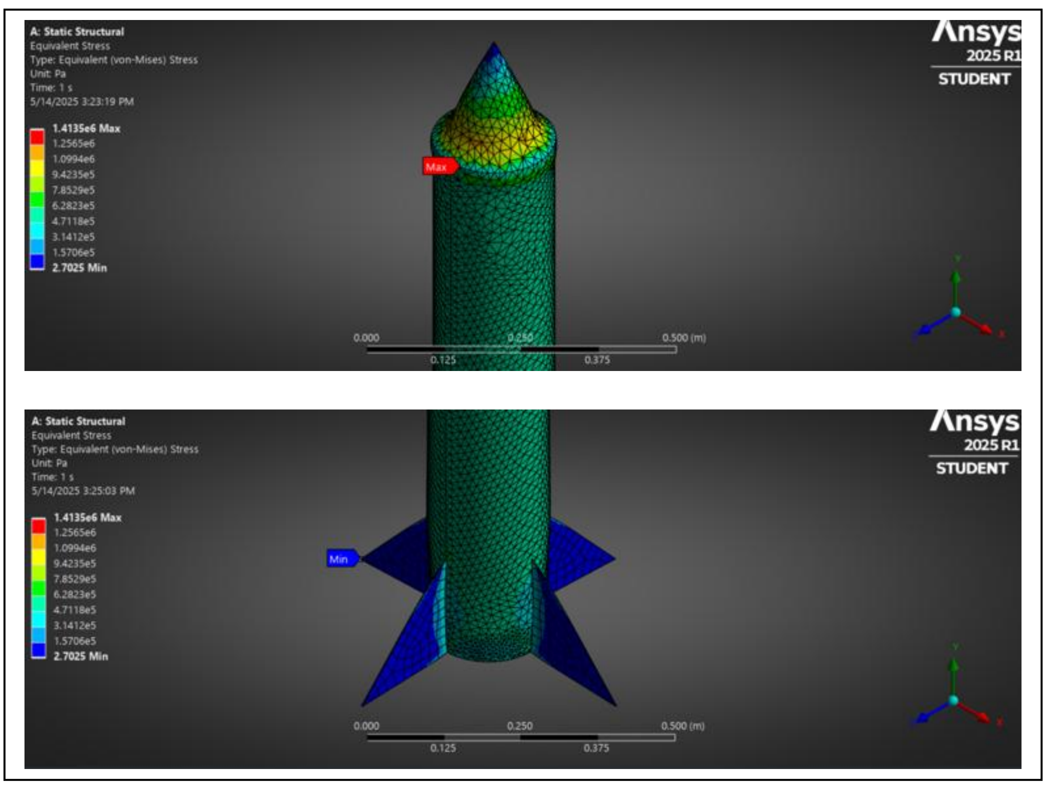


Figure 17: Max (top) and Min (bottom) Stress of the Titanium Rocket

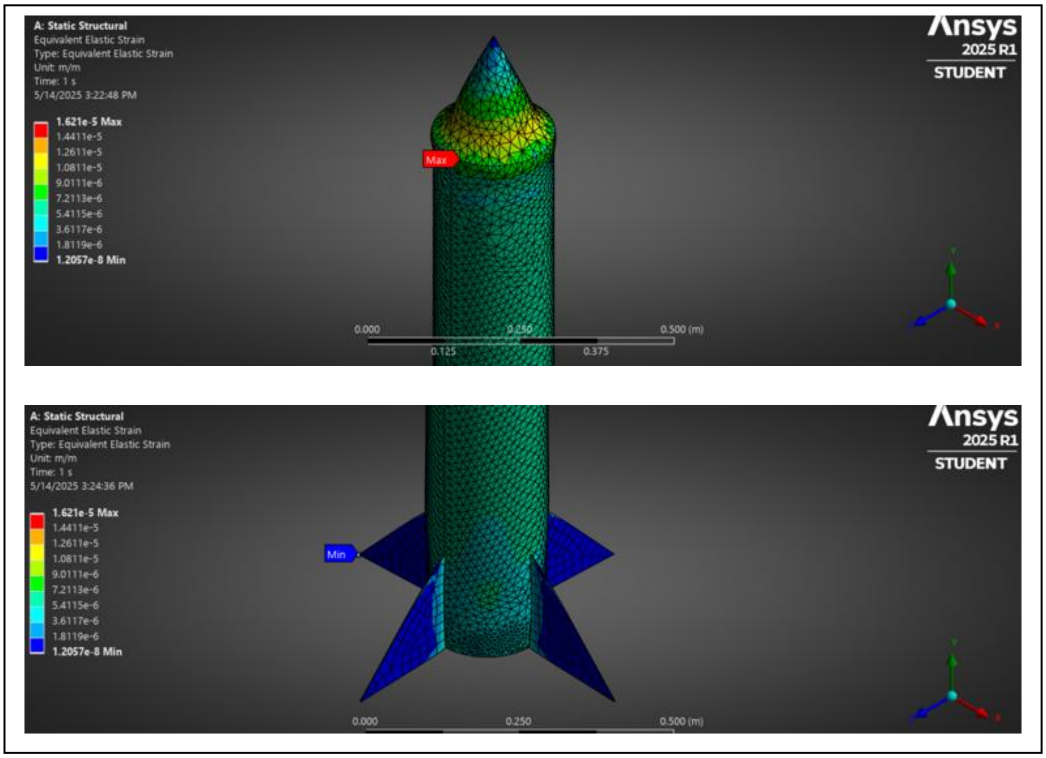


Figure 18: Max (top) and Min (bottom) Strain of the Titanium Rocket

3. **Carbon Fiber (395GPa)**

Now, the same simulation was run on a non-metal composite material which was the toughest carbon fiber in ANSYS's material directory. This was done to see how a structure based on a non-metal composite material would act and deform under the same conditions as an identical metallic structure. The results were significantly different in this case. The maximum deformation (Figure 19), stress (Figure 20), and strain (Figure 21) faced by the carbon fiber structure was 2.5191e-4 meters, 3.5694e6 Pascals, and 5.3942e-4 respectively, while the minimum values were very insignificant (compared to the maximum values) or completely 0.

The maximum strain and stress values are very interesting due to the fact that they exist on the point where the fin connects to the main body tube. The simulation also signifies that the composite fins are considerably weaker than their metallic counterparts. The maximum stress faced by carbon fiber is about 152.52% more than its titanium twin, which is likely due to the rigid and non-malleable nature of composite materials. The material deforms significantly more (around 1482.95%) than the same structure made from titanium! This much deformation can cause catastrophic failures in the structure if such rockets are flown.

This is probably the reason why the use of carbon fiber is not widespread in the construction of vehicles/equipment that experience extreme stresses even if the material is lighter. The limitation is that the material is not very strong under compressive loads when compared to tensile loads.

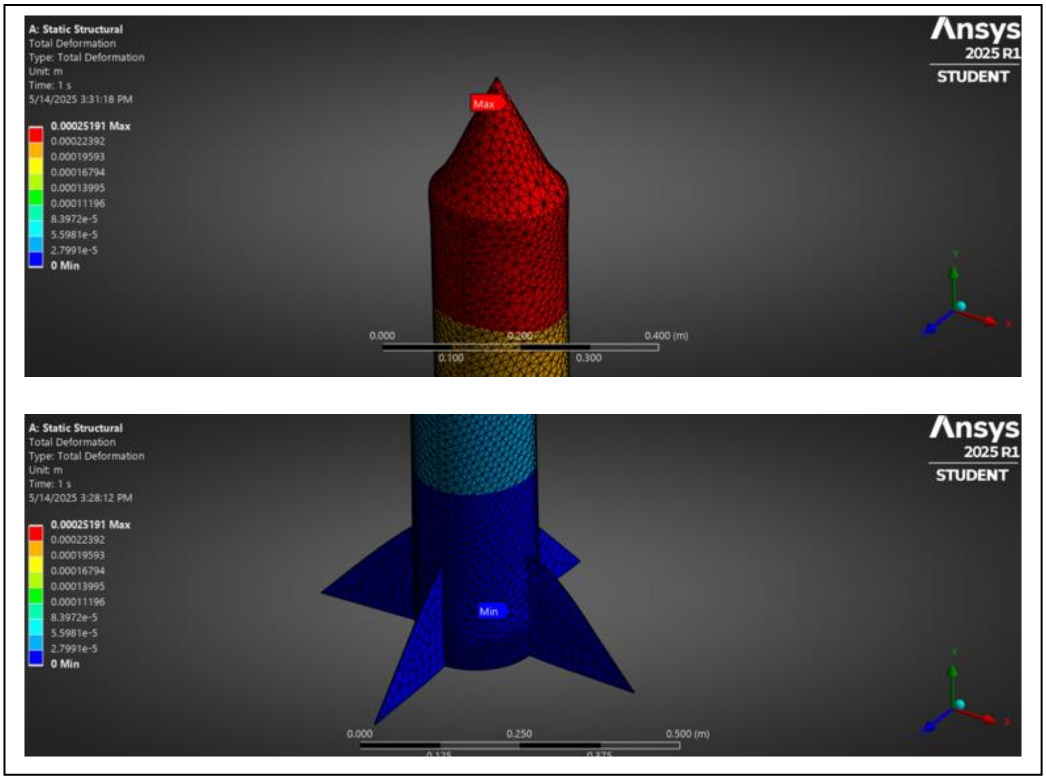


Figure 19: Max (top) and Min (bottom) Deformation of the Carbon Fiber Rocket

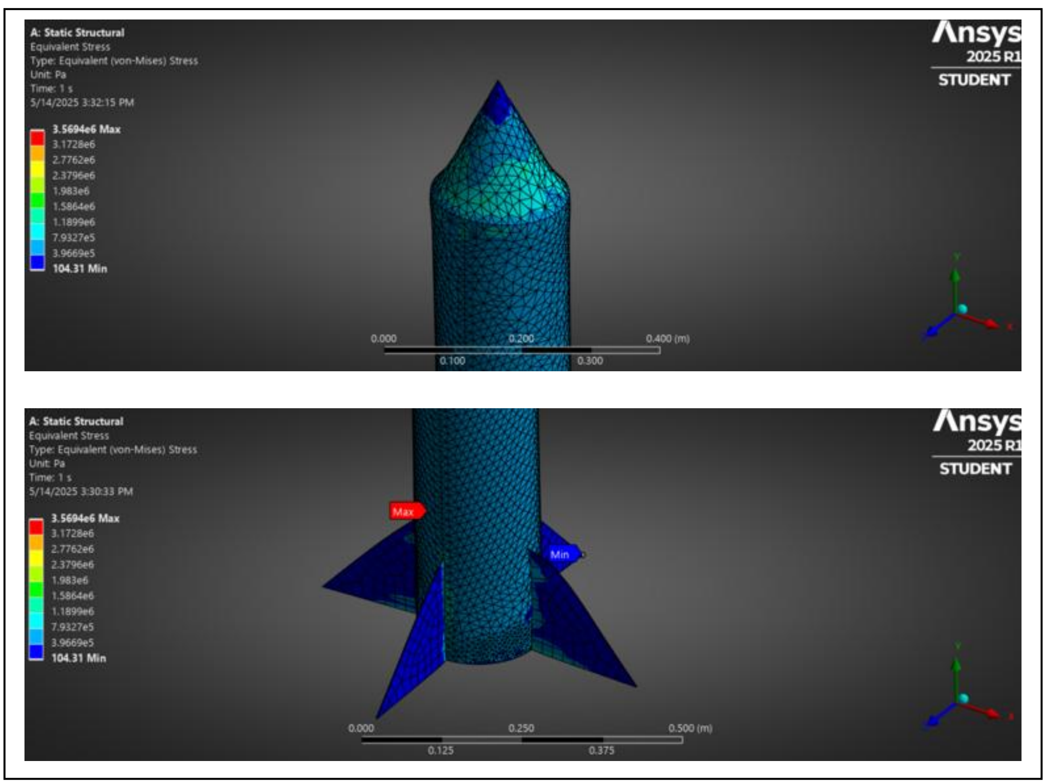


Figure 20: Max/Min (bottom picture) Stress of the Carbon Fiber Rocket

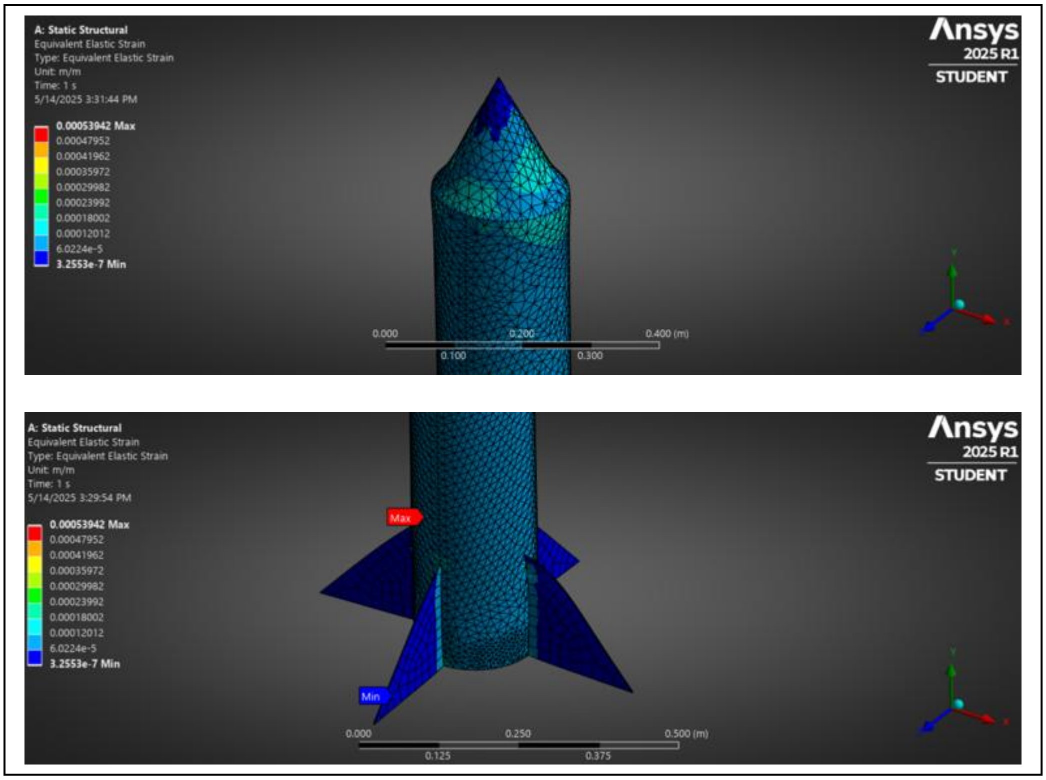


Figure 21: Max/Min (bottom picture) Strain of the Carbon Fiber Rocket

Discussion

Effects due to increased thickness

One of the main changes in the CAD model was the increase in wall thickness from 0.08mm to 0.2mm. This was done because a thickness of 0.08mm was too thin for Mechanical software to mesh appropriately, resulting in an increase of roughly 138.95%. This implies that the deformation would be worse, as deformation is inversely related to the cross sectional area of a body according to Eq. (2), where δ is the wall thickness. A reduced surface area would increase the deformation by a factor of about 2.5, and so would the equivalent stress. Equation 3 shows the relation between the applied force and the cross sectional area.

$$\delta = \frac{FL}{A_{CS}E} \qquad \textbf{Eq. (2)}$$

$$\sigma = \frac{F}{A_{CS}} \qquad \textbf{Eq. (3)}$$

Impacts due to Material Properties

Metal alloys like aluminum and titanium are appropriate materials to construct structures that are expected to experience high compressive loads. Their predictable behavior allows for safe construction of such structures.

Titanium alloy performed better than aluminum alloy due to its increased hardness and density. This resulted in a reduction of deformation experienced by the rocket by about 26% under roughly the same amount of stress. However, titanium is almost twice as dense as aluminum, which would mean that the rocket would be double its original weight, making it not very ideal for launching adequately.

Composite materials like carbon fiber are excellent for structures that are expected to experience high tensile loads. However, a rocket structure is expected to experience high compressive loads, which is not a suited field of application for such a material. The results from ANSYS Static Structural more or less confirm this by showing that the deformation of the carbon fiber structure was increased by 1482.95% compared to its titanium counterpart. The fins made from carbon fiber also experience higher buckling, making them significantly weaker than the metal ones.

When the actual thickness of 0.08mm is considered, the aluminum and titanium structures will most likely have a deformation of about 5.511e-5 meters and 3.9785e-5 meters respectively (applying the scale factor of 2.5), which is not a significant deformation, so the rocket might actually survive if it is fastened adequately! Although, the same cannot be concluded for the rocket made from carbon fiber, which will deform more than the previous case if the thickness is reduced.

Verification

The results were verified by changing the mesh size from 0.02m to 0.025m (25% increase) for the aluminum rocket. If the maximum values had a percentage difference amongst themselves less than 25%, the result would be considered adequate. The percentage difference was found to be 3.25e-3%, 17.86%, and 23.62% for deformation, stress, and strain respectively. These values are well within the verification bound, so the simulation was considered to be accurate. Similar results were then assumed for the titanium and carbon fiber rockets, whose “verification” simulations were not performed then.

The results for the verification simulation of the aluminum rocket are shown in Figures 22, 23, and 24 on the next page.

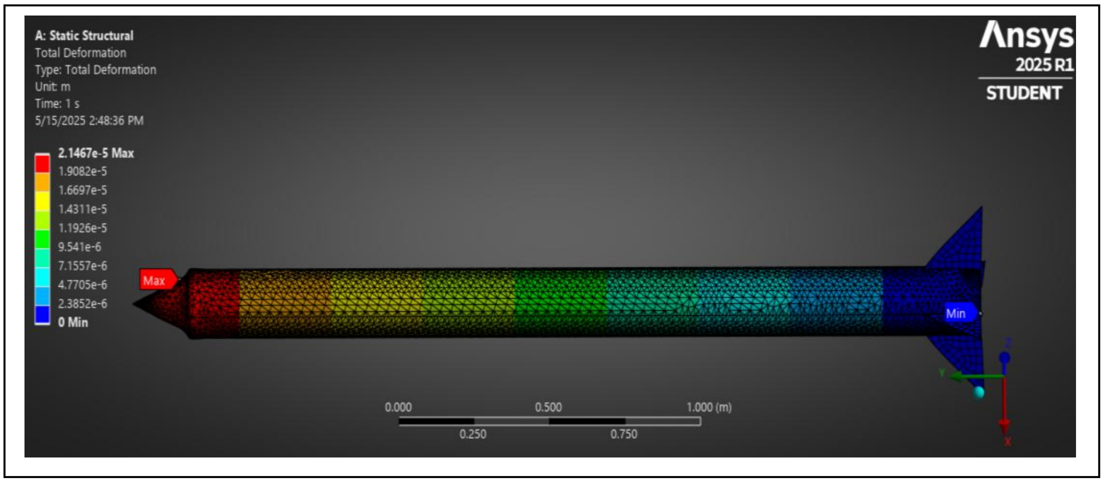


Figure 22: Max/Min Deformation of the Carbon Fiber Rocket (Verification)

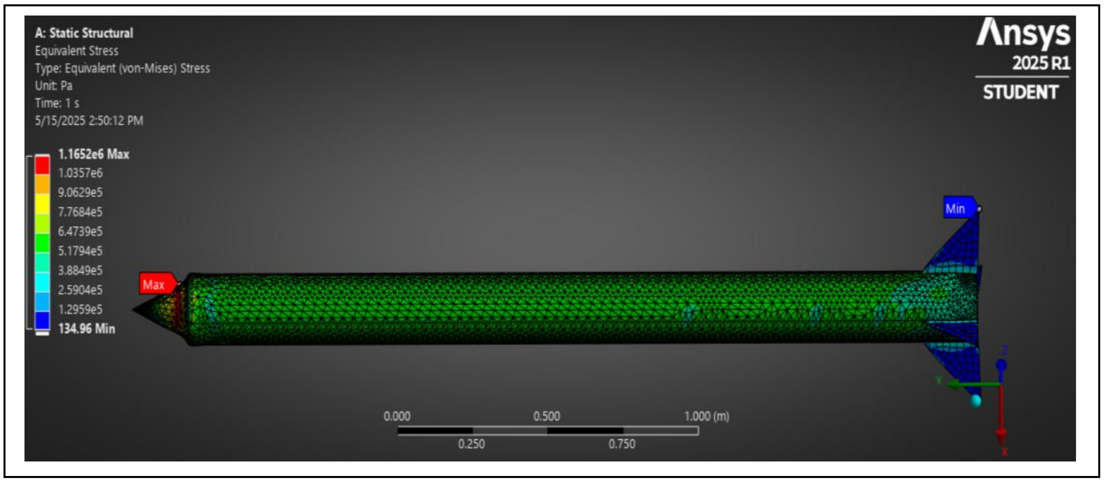


Figure 23: Max/Min Stress of the Carbon Fiber Rocket (Verification)

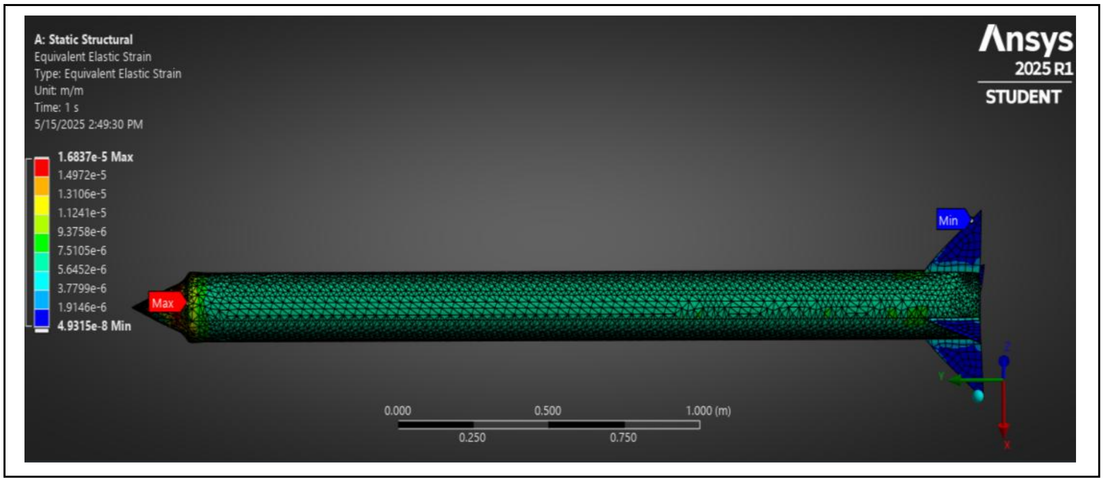


Figure 24: Max/Min Strain of the Carbon Fiber Rocket (Verification)

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11. *Students*. (2025, April 17). https://www.ansys.com/academic/students

Appendix A

```
%MAE 423 - Project Code
%Humza Khan (50363412)
clc; clear all; close all;

%% project starts here
%inputs
g = 9.81; %m/s
payload = [5, 10, 20]; %kg
h_max = [10000; 20000; 30000] / 3.281; %altitude converted to meters
a_max = [5, 10, 20]; %normalized acc
all_combos = [payload(1), h_max(2), a_max(2);
              payload(3), h_max(2), a_max(2);
              payload(2), h_max(1), a_max(2);
              payload(2), h_max(3), a_max(2);
              payload(2), h_max(2), a_max(1);
              payload(2), h_max(2), a_max(3);
              payload(2), h_max(2), a_max(2)]; %all possible inputs

SM = 2; %static margin
rho_s = 2700; %kg/m^3
rho_p = 1772; %kg/m^3
sigma_s = 150e6; %MPa ---> Pa
N = 4; %Number of Fins
R_opt = zeros(7,1); %initializing variable
W_eq = zeros(7,1); %initializing variable
t_burn = zeros(7,1); %initializing variable
mach = zeros(7,1); %initializing variable
[T_sl,a_sl,P_sl,rho_sl] = atmosisa(0);
T_stp = 298.15; %K Temp at STP
gamma = 1.4; %assuming diatomic atmosphere
R = 287; %universal gas constant
a_0 = sqrt(gamma.*T_stp.*R);

%============================================================================
%% Solving for R_opt, W_eq, t_b, and Mach Numbers for each case
for i = 1:7
    R_opt(i) = 1 + all_combos(i,3);
    W_eq(i) = sqrt((all_combos(i,2).*g)./(0.5.*log(R_opt(i)).*(log(R_opt(i))-2)+((R_opt(i) - 1)./
R_opt(i))));
    t_burn(i) = ((R_opt(i) - 1)*W_eq(i))/(g*R_opt(i)); %optimal burn time eqn
    mach(i) = W_eq(i)/a_0; %solving for mach numbers @ SL
end
%============================================================================
%% solving for gas dynamics stuff incl. total temp and pressure, mass, cg, cp
 T_1 = T_stp.*(1+0.5.*(gamma-1).*mach.^2) ; %K total temp
 P_0 = P_sl.*(T_1./T_stp).^(gamma/(gamma-1)) ; %Pa total pressure

guess_d = linspace(0,1,5000); %guess D range
guess_L = linspace(0,5,5000); %guess L range
%============================================================================
%starting the loop
for k = 1:7
    delta = guess_d .* P_0(k,1) ./ (2 * sigma_s);
    L_D = guess_d + guess_L';
```

```
%==========================================================================
    %mass calculations
    %fin
    fin_sa = 0.5 .* guess_d .* guess_d; %SA of fin
    m_fin = N .* fin_sa .* delta .* rho_s; %mass of 4 fins
    m_fin_matrix = repmat(m_fin,[5000,1]); %matrix sizing
%==========================================================================
    %cone
    cone_sa = pi.*(0.5.*guess_d).*(sqrt( (0.5.*guess_d).^2 + guess_d.^2)); %SA of cone (no bottom
circle)
    m_cone = cone_sa .* delta .* rho_s; %mass of the cone
    m_cone_matrix = repmat(m_cone,[5000,1]);%matrix sizing
%==========================================================================
    %cylinder
    cylinder_sa = 2.*pi.*(L_D).*(0.5.*guess_d); %SA of cylinder (no top/bottom circle)
    m_cylinder = cylinder_sa .* delta .* rho_s; %no need to adjust sizing as dependant on both L
and D.
%==========================================================================
    %full structure
    m_structure = m_fin_matrix + m_cone_matrix + m_cylinder;
%==========================================================================
    %propellant mass
    m_p = (R_opt(k,1)-1) .* (m_structure + all_combos(k,1));
%==========================================================================
    %propellant length
    L_p = m_p ./ (guess_d.^2 .* (pi.*rho_p/4));
%==========================================================================
   %X_CG Calculation
   X_CG_nose = (2./3).*guess_d;
   X_CG_cylinder = guess_d + 0.5.*(L_D);
   X_CG_fins = (2./3).*guess_d + (L_D);
   X_CG_prop = (0.5.*L_p) + ((2.*guess_d + guess_L') - L_p);
   m_total = m_structure + m_p + all_combos(k,1);
   X_CG = ((X_CG_fins.*m_fin_matrix) + (X_CG_cylinder.*m_cylinder) + (X_CG_prop.*m_p) +
(X_CG_nose.*m_cone_matrix) + (X_CG_nose .* all_combos(k,1)))./m_total ;
%==========================================================================
    % X_CP calculation - Most Values were obtained from the Centuri Document
    X_conical = (2./3).*guess_d; %recalculating for sanity %center for normal force
    X_conical_matrix = repmat(X_conical,[5000,1]);
    a = guess_d;
    b = 0;
    S = guess_d;
    weird_l_length = sqrt(guess_d.^2 + (0.5.*guess_d).^2);
    CNaf = (4.*N.*(S./guess_d).^2)./(1 + sqrt(1 + (2.*weird_l_length./(a + b)).^2)); %found this
to be constant throughout
    R = 0.5.*guess_d; %radius
    kfb = 1 + (R./(S+R)); %interferance factor for 4 fins
    CNafb = kfb.*CNaf; %normal force on fins in presence of the body
    CNan = 2; %normal force on the nose
    x_f = guess_d + guess_L';
    m = guess_d;
    del_xf = (((m.*(a + 2.*b))./(3.*(a + b)))) + (1/6).*(a + b - (a.*b)./(a+b));
    xf_bar = x_f + repmat(del_xf,[5000,1]); %total center of pressure in inches
    CNa = CNafb + CNan; %total norm coeff
    %solving for x_cp
    X_CP = (CNan.*X_conical_matrix + repmat(CNafb,[5000,1]).*xf_bar)./CNa; %center of pressure
location from the top of the rocket
```

```
%==========================================================================
    lamda = all_combos(k,1)./(m_structure+m_p);

    %stability analysis
    const_1 = abs((X_CP - X_CG) ./ repmat((guess_d),[5000,1]));
    const_1norm = abs(const_1) - 2; %enforces SM normalizes
    const_1norm1 = abs(const_1norm);

    min_const1(1,k) = min(const_1norm1(:));
    [min_row, min_col] = find(const_1norm1 == min_const1(1,k));

    const_2 = lamda;

    max_lamda(1,k) = const_2(min_row, min_col);
    [max_row, max_col] = find(const_2 == max_lamda(1,k));

%==========================================================================
    %% Extracting Case-By-Case Values in this section
    P0_fraction(1,k) = P_0(k,1)/P_sl;
    dD_fraction(1,k) = delta(1,max_col)/guess_d(1,max_col);
    final_d(1,k) = guess_d(1,max_col);
    final_L(1,k) = guess_L(1,max_row);
    LD_fraction(1,k) = final_L(1,k)/final_d(1,k);
    final_XCG(1,k) = X_CG(max_row,max_col);
    final_XCP(1,k) = X_CP(max_row,max_col);
    final_mp(1,k) = m_p(max_row,max_col);
    final_ms(1,k) = m_structure(max_row,max_col);
    final_m0(1,k) = m_total(max_row,max_col);
    final_lamda(1,k) = max_lamda(1,k);
    epsilon(1,k) = m_structure(max_row,max_col)/(m_structure(max_row,max_col) +
m_p(max_row,max_col));

%==========================================================================

end
%thrust = m_p.*W_eq./t_burn; ----> Thrust eqn used.
%==========================================================================

%% organizing data in arrays
Cases = all_combos;
dataset_extract = [R_opt, W_eq, t_burn, P0_fraction', dD_fraction',final_d', final_L',
LD_fraction', final_XCG', final_XCP', final_mp', final_ms', final_m0', final_lamda',epsilon'];
FINAL_TABLE = [Cases, dataset_extract]; %final answer that was used to form the table.
%==========================================================================
```